Cross-Dataset Generalisation and Decision Behaviour of Machine Learning Models for Fake News Detection

Chapters 1–3 (Revised Draft)

CHAPTER 1: INTRODUCTION AND OVERVIEW

1.1 Introduction

The rapid growth of digital media and social networking platforms has significantly changed how people access, share and consume news. News information can now be distributed to large audiences within a very short period of time, allowing important information to reach the public quickly. However, the same communication environment also enables false or misleading information to spread rapidly. The scale and speed of online information make manual verification difficult to maintain as the only method of identifying false news at scale. Fact-checking organisations provide valuable expert verification, but human-based verification is labour-intensive and cannot realistically examine every piece of news content produced and shared online. The development of automated methods for identifying potentially misleading news has therefore become an important research problem within Natural Language Processing (NLP), machine learning and artificial intelligence.

Automated fake news detection can be formulated as a classification problem in which computational models analyse characteristics of news content and predict whether an article belongs to a real or fake news class. Previous research has demonstrated that textual characteristics can provide useful signals for this task. Linguistic patterns associated with misinformation may include emotional or sensational language, informality, subjectivity and other stylistic characteristics. Shu et al. (2017) identify several perspectives for fake news detection, including content and social-context features, while Zhou and Zafarani (2020) categorise detection approaches into knowledge-based, style-based, propagation-based and source-based methods.

Machine learning and NLP techniques have consequently been widely applied to automated fake news detection. Traditional approaches can transform textual information into numerical representations such as Term Frequency-Inverse Document Frequency (TF-IDF), which can then be used by supervised classification algorithms. The original research undertaken for this project investigated this approach using the ISOT Fake News Dataset (Ahmed, Traore and Saad, 2017; 2018) and three supervised classifiers: Naive Bayes, Logistic Regression and Random Forest. The dataset contains approximately 44,898 labelled news articles and was selected because of its transparent construction and suitability for supervised classification.

However, achieving strong classification performance on a benchmark dataset does not necessarily demonstrate that a model will perform reliably when it encounters news originating from a different dataset, domain, source or distribution. This issue is particularly important for fake news detection because datasets can differ in their sources, topics, publication periods, labelling procedures and linguistic characteristics. The original research already recognised that the ISOT dataset is predominantly based on English-language news from a specific period and may therefore contain cultural and temporal limitations affecting generalisability.

Recent research has increasingly investigated cross-domain and cross-dataset generalisation rather than relying exclusively on conventional benchmark evaluation. Janicka, Pszona and Wawer (2019) provided early empirical evidence for this problem, showing that classifiers trained on one fake news dataset and tested on another can lose more than twenty percentage points of accuracy. A recent large-scale benchmarking study spanning ten datasets and twelve model families reported a comparable pattern: fine-tuned transformer models achieved a median F1 score of 0.86 when evaluated in-domain but fell to a median of 0.53 under cross-dataset evaluation (Dell'Oglio et al., 2026). This body of evidence suggests that the ability of a model to generalise beyond the dataset on which it was trained is an important consideration when assessing the reliability of automated fake news detection systems, and that the generalisation gap can be substantial rather than marginal. At the same time, explainability techniques provide an opportunity to investigate the factors influencing individual model predictions rather than evaluating a model solely through aggregate performance metrics.

This research therefore extends the original project from a conventional model comparison into an investigation of model generalisation and decision behaviour under dataset shift. Instead of asking only which model achieves the highest score on a benchmark dataset, the research investigates how model performance changes when the model is evaluated on data from a different distribution and what characteristics influence its predictions under these conditions.

1.2 Research Problem

The central problem addressed by this research is that high performance on a benchmark fake news dataset may provide an incomplete indication of how reliably a machine learning model will perform on unseen news from a different dataset or domain.

The original project evaluated three supervised classifiers under controlled conditions using the ISOT dataset. Its research questions focused on NLP preprocessing, comparison of Naive Bayes, Logistic Regression and Random Forest, and identification of predictive linguistic features. While this approach provides a useful baseline for automated fake news classification, it primarily evaluates performance within a single benchmark environment.

A model can potentially learn patterns that are strongly associated with the dataset used during training rather than patterns that represent general characteristics of misinformation. Differences in publisher sources, topics, writing styles, publication periods and dataset construction can therefore affect model performance when the model is applied outside its original training distribution. Janicka, Pszona and Wawer (2019) attribute this directly to dataset-specific lexical and stylistic artefacts rather than to genuine indicators of misinformation, which is the concern this dissertation investigates empirically.

This creates an important research problem. A model achieving high accuracy on its benchmark test set may appear highly effective even when its performance decreases substantially on an unseen dataset. Conventional metrics such as accuracy and F1 score can identify that performance has changed, but they do not necessarily explain why the change has occurred or which features have influenced the model's decisions.

The research therefore addresses two connected issues:

Generalisation: whether fake news detection models maintain their predictive performance when evaluated on a dataset different from the one used for training.

Model behaviour: whether explainability and systematic error analysis can provide evidence about the textual or dataset-specific characteristics influencing model predictions under both in-domain and cross-dataset conditions.

The purpose of this research is not therefore simply to identify the most accurate classifier. Instead, it investigates the reliability and behaviour of fake news detection models when they are exposed to changes in the data distribution.

1.3 Research Context

Fake news detection has developed from relatively simple content-based classification approaches towards increasingly sophisticated machine learning and deep learning systems. Early research demonstrated that linguistic and content-based characteristics could provide useful signals for distinguishing fake and legitimate news. More recent approaches have incorporated contextual representations, transformer-based architectures such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), external evidence and explainability.

The original project focused on content-based detection because textual information provides a practical and reproducible source of information for automated classification. The ISOT dataset was selected as the primary dataset and contains separate collections of real and fake articles. The original methodology proposed combining text preprocessing and TF-IDF feature extraction with supervised classifiers.

However, a practical fake news detection system should ideally operate beyond the exact distribution represented by its training data. Real-world news is produced by different publishers and across different topics, periods and writing styles. Consequently, evaluating a model only on randomly held-out examples from the same dataset may not provide sufficient evidence of its robustness.

Cross-dataset evaluation provides a more demanding experimental setting. A model trained on one dataset is evaluated against another dataset that was not used during training. The resulting difference between in-domain and cross-dataset performance can therefore provide evidence about the extent to which the model generalises.

Explainability provides a second dimension to this investigation. Rather than treating the model as a system that only produces a final classification, explainability techniques such as LIME (Ribeiro, Singh and Guestrin, 2016) and SHAP (Lundberg and Lee, 2017) can be used to examine the features contributing to individual predictions. When combined with error analysis, this can help investigate whether models are relying on meaningful linguistic patterns or potentially dataset-specific characteristics — a phenomenon described in the wider machine learning literature as shortcut learning (Geirhos et al., 2020).

The research therefore combines classification performance, cross-dataset evaluation, explainability and error analysis into a single experimental framework.

1.4 Research Gap

The literature demonstrates that machine learning and NLP can achieve strong results for fake news detection, and research has also increasingly explored more advanced models, cross-domain evaluation and explainable approaches. However, strong benchmark performance alone does not establish that a model has learned features that generalise reliably beyond the data distribution on which it was trained.

The original project addressed transparency by using an openly available benchmark dataset and interpretable machine learning algorithms. It also recognised the limitation that the ISOT dataset represents a particular temporal and linguistic distribution.

The revised research addresses this limitation by moving the focus from model performance within one dataset to model behaviour across datasets. This is not an unstudied area — Janicka, Pszona and Wawer (2019) and, more recently, Dell'Oglio et al. (2026) have both demonstrated substantial cross-dataset performance degradation across a range of model families. However, neither study combines cross-dataset evaluation with the fine-grained, feature-level explainability and manual error analysis proposed here, and neither is built specifically around the ISOT dataset and the traditional TF-IDF pipeline established in the original project.

The research gap investigated in this project is therefore:

There is a need for empirical investigation into how fake news detection models generalise across different datasets and how explainability and error analysis can be used to understand changes in their decision behaviour under dataset shift.

This research does not claim that cross-dataset fake news detection or explainable AI has never been investigated. Instead, its contribution is based on designing a controlled experimental investigation that connects in-domain performance, cross-dataset performance, feature attribution and error analysis to examine the reliability of model predictions, using the ISOT-based pipeline established in the original project as its foundation.

This distinction is important because a decrease in performance between datasets is itself only an observation. The research seeks to go further by investigating the possible factors associated with that decrease.

1.5 Research Aim

The aim of this research is:

To investigate the cross-dataset generalisation and decision behaviour of machine learning models for fake news detection, using explainability and error analysis to identify features and potential dataset-specific patterns that influence model predictions.

1.6 Research Questions

The research is guided by the following research questions:

RQ1 — How does the performance of fake news detection models change when models trained on one benchmark dataset are evaluated on a different dataset?

RQ2 — What linguistic and dataset-specific features influence fake news detection model predictions under in-domain and cross-dataset evaluation?

RQ3 — How can explainability and systematic error analysis be used to identify potential shortcut learning and differences in model behaviour under dataset shift?

These questions extend the original research questions, which focused primarily on applying NLP and TF-IDF, comparing three classifiers on ISOT, and identifying predictive vocabulary features.

1.7 Research Objectives

To achieve the research aim, the following objectives have been established:

Objective 1 — To develop a reproducible NLP-based preprocessing and machine learning pipeline for binary fake news classification.

Objective 2 — To establish baseline performance using selected machine learning models on the ISOT Fake News Dataset.

Objective 3 — To evaluate the selected models under cross-dataset conditions using the WELFake dataset (Verma et al., 2021) as an unseen secondary dataset.

Objective 4 — To quantify differences between in-domain and cross-dataset model performance using appropriate classification metrics and significance testing.

Objective 5 — To apply explainability techniques (LIME and SHAP) to investigate the features influencing model predictions under different evaluation conditions.

Objective 6 — To conduct systematic error analysis to identify potential dataset-specific patterns and investigate the circumstances in which models fail to generalise.

1.8 Research Contribution

The contribution of this research is not intended to be the development of another fake news classifier based solely on achieving a higher classification score. Instead, the research contributes an experimental framework for investigating how reliable and generalisable fake news detection models are when the evaluation data differs from the training data.

The study will contribute in four main areas.

First, it will provide a controlled comparison between in-domain and cross-dataset performance, allowing the degree of performance degradation under dataset shift to be quantified using ISOT as the training distribution and WELFake as the unseen evaluation distribution.

Second, it will investigate model decision behaviour using explainability techniques, providing information about the textual characteristics contributing to predictions.

Third, it will combine explainability with systematic error analysis to investigate potential dataset-specific patterns or shortcut behaviour that may not be visible from conventional performance metrics alone.

Finally, the research will implement these methods within a practical fake news detection artefact — a lightweight interactive web application in which a user submits news text and receives a model prediction together with a visual explanation of the words that most influenced it — allowing users to inspect the factors contributing to a prediction rather than receiving only a binary label.

The contribution is therefore both research-oriented and practical: the research investigates model generalisation and behaviour, while the artefact demonstrates the application of the resulting approach in an interactive system.

1.9 Scope of the Research

The research focuses on text-based binary fake news detection using publicly available datasets. The primary research foundation is the ISOT Fake News Dataset used in the original project. The secondary dataset used for cross-dataset evaluation is WELFake (Verma et al., 2021), a corpus of 72,134 labelled articles constructed by merging four independently collected sources (Kaggle, McIntire, Reuters and BuzzFeed Political). WELFake was selected over alternatives such as LIAR (Wang, 2017) because its article-level format and binary real/fake labelling are directly compatible with ISOT, whereas LIAR consists of short political statements with a six-class truthfulness scale that would require substantial label remapping and would introduce a confounding change in text genre alongside the change in dataset. This choice is treated as provisional pending the dataset inspection described in Chapter 3, and an alternative will be substituted if inspection reveals unresolvable compatibility issues.

The research will investigate traditional machine learning models (Naive Bayes, Logistic Regression, Random Forest) using TF-IDF representation, and a transformer-based model (RoBERTa-base; Liu et al., 2019) as a contextual comparison, continuing the direction identified as future work in the original project. This selection is likewise treated as a working default, to be confirmed or revised following the feasibility testing described in Chapter 3.

The study will evaluate models using standard classification metrics, including accuracy, precision, recall, F1 score and, where appropriate, ROC-AUC. These metrics were also used in the original project.

The research will also incorporate explainability (LIME and SHAP) and systematic error analysis to investigate model behaviour.

The project does not attempt to establish the factual truth of arbitrary real-world claims independently through human fact-checking. Instead, it investigates the behaviour and generalisation of machine learning models trained on labelled datasets. Consequently, predictions produced by the artefact should be understood as model classifications rather than definitive statements of factual truth.

1.10 Proposed Research Approach

The research adopts a quantitative, experimental computational approach. The original project was positioned using a positivist philosophy and deductive approach, with controlled computational experimentation and numerical performance measures. The revised study retains this quantitative experimental foundation while expanding the experimental design to investigate generalisation and model behaviour.

The proposed research process consists of the following stages: dataset preparation, preprocessing, feature representation, baseline model development, in-domain evaluation, cross-dataset evaluation, generalisation analysis, explainability analysis, error analysis and artefact development. The full stage-by-stage pipeline is set out in Section 3.27.

This approach ensures that the artefact is supported by the experimental research rather than functioning as an independent software-development exercise.

1.11 Significance of the Research

Reliable fake news detection is important because automated systems may be used to support decisions about information that users encounter online. However, a system that performs well only on a particular benchmark dataset may provide misleading confidence when applied to different forms of news, as the scale of the generalisation gap reported by Dell'Oglio et al. (2026) illustrates.

Understanding model generalisation is therefore important for evaluating the practical reliability of fake news detection systems.

The research is also significant because classification performance alone cannot explain how a model reaches its decisions. By incorporating explainability and error analysis, the study aims to provide a deeper understanding of the behaviour of the models rather than treating accuracy as the only indicator of success.

This is particularly relevant to the development of responsible AI systems. The previous project already recognised that false-positive predictions could incorrectly label legitimate news as fake and therefore should not be interpreted as a basis for automated censorship. The revised research maintains this principle and treats the resulting system as a decision-support and research artefact, rather than an autonomous authority on factual truth.

1.12 Structure of the Applied Research Project

The remainder of this Applied Research Project is organised into several sections.

Chapter 2 presents the critical literature review, examining research in automated fake news detection, NLP-based classification, benchmark datasets, model generalisation, explainable AI and dataset-specific limitations. The chapter critically evaluates previous research to establish the basis for the research gap addressed by this study.

Chapter 3 presents the methodology and research design, including dataset selection, preprocessing, feature representation, model selection, experimental design, in-domain and cross-dataset evaluation, explainability techniques, evaluation metrics and ethical considerations.

Chapter 4 presents the development, experimental results and analysis, including implementation of the detection pipeline, model performance, cross-dataset results, explainability outputs and systematic error analysis. The practical artefact developed as part of the research is also presented.

Chapter 5 presents the conclusions and future work, summarising the findings in relation to the research questions, discussing the research contribution and limitations, and identifying opportunities for further research.

1.13 Chapter Summary

This chapter introduced the research problem of automated fake news detection and established the limitations of evaluating machine learning models solely through performance on a single benchmark dataset. Evidence from Janicka, Pszona and Wawer (2019) and Dell'Oglio et al. (2026) demonstrated that this limitation is not merely theoretical: cross-dataset performance drops of twenty percentage points or more have been documented empirically. The chapter identified cross-dataset generalisation and model decision behaviour as the central focus of the revised research.

The research aim, three research questions and six research objectives were defined. The proposed contribution is to investigate the relationship between in-domain performance, cross-dataset generalisation, explainability and error behaviour rather than simply comparing models according to their benchmark accuracy.

The research will use a quantitative experimental approach and will build on the ISOT-based work completed in the previous stage of the project, extending it with the WELFake dataset for cross-dataset evaluation. The next chapter will critically examine the existing literature and establish the evidence required to support the research gap and experimental design.

CHAPTER 2: CRITICAL LITERATURE REVIEW

2.1 Introduction

Fake news detection has developed into a multidisciplinary research area involving Natural Language Processing (NLP), machine learning, data mining, social computing and information retrieval. The growth of online news platforms and social media has increased the speed at which misleading information can be produced and distributed, creating a need for computational methods capable of supporting the identification of potentially false information at scale.

The previous stage of this research established a foundation for text-based fake news detection using supervised machine learning, examining the ISOT Fake News Dataset (Ahmed, Traore and Saad, 2017; 2018) and investigating Naive Bayes, Logistic Regression and Random Forest using NLP preprocessing and TF-IDF feature representation. The literature review supporting that work identified fake news as a problem that can be approached through content, style, knowledge, propagation and source-based information.

However, the objective of the present research is broader than determining which classifier performs best on a single benchmark dataset. The literature increasingly raises questions about dataset limitations, generalisability, interpretability and the extent to which machine learning models learn meaningful characteristics of misinformation rather than characteristics specific to the datasets on which they are trained.

This chapter therefore critically reviews the literature relevant to the revised research direction. It first establishes the conceptual foundations of fake news detection, before examining content-based and style-based detection, traditional machine learning, transformer-based approaches, benchmark datasets, cross-dataset generalisation, explainability and shortcut learning. The chapter then critically evaluates the limitations identified across these areas and develops the research gap addressed by this study.

2.2 Understanding Fake News Detection

One of the foundational works reviewed for this research is Shu et al. (2017), who present fake news detection from a data-mining perspective. Their framework treats detection as a computational classification problem involving feature extraction followed by model construction. Importantly, the authors distinguish between news content features and social context features. Content features include linguistic and semantic characteristics of the news itself, whereas social context features capture information relating to users, engagement and the way information propagates through social networks.

This distinction is important because it demonstrates that fake news is not exclusively a textual phenomenon. A news article may contain relatively credible language while its propagation behaviour, source or social context provides additional evidence about its credibility. Consequently, a purely text-based classifier operates on only one part of the information available in a real-world misinformation environment.

For the present research, however, textual analysis remains the primary focus. This provides a controlled environment in which the behaviour of different NLP models can be investigated across datasets without introducing the additional complexity of social-network data. The decision is therefore methodological rather than a claim that textual information is sufficient for complete fact verification.

Shu et al. (2017) also highlight the difficulty created by the fact that fake news can intentionally imitate credible journalism. Clickbait headlines, emotionally charged vocabulary and persuasive writing can make false information resemble legitimate reporting. This creates a fundamental challenge for content-based detection: features that appear predictive within a particular dataset may not necessarily represent the factual reliability of the underlying information.

This observation becomes particularly relevant to the current research because a model may learn the style associated with a dataset’s fake-news examples rather than learning general indicators of misinformation — the shortcut-learning concern developed further in Section 2.10.

2.3 Theoretical Perspectives on Fake News Detection

Zhou and Zafarani (2020) provide a broader theoretical framework by organising fake news detection into four major perspectives: knowledge-based, style-based, propagation-based and source-based detection.

Knowledge-based approaches attempt to determine whether claims contained within news content are consistent with verified information. This can involve comparing extracted factual relationships against knowledge graphs, fact-checking databases or other trusted information sources. The major advantage of this approach is that it moves beyond linguistic appearance, but such systems introduce significant requirements for external knowledge, evidence retrieval and factual verification, and can encounter difficulties when claims are new, ambiguous, incomplete or unavailable in existing knowledge sources.

Style-based approaches analyse characteristics of the language used in an article, including vocabulary, emotional language, subjectivity, informality, lexical characteristics and syntactic patterns. This approach is particularly relevant to the present research because the original project used TF-IDF and supervised classifiers to investigate textual patterns. Literature reviewed by Zhou and Zafarani (2020) indicates that fake news can exhibit higher levels of informality, emotional language, subjectivity and lexical diversity than legitimate reporting. However, style-based detection has an important weakness: writing style is not equivalent to factual truth. A legitimate article can contain emotional language, while a professionally written false article can deliberately imitate legitimate journalistic style. A classifier that relies heavily on stylistic patterns may therefore achieve high benchmark performance without necessarily learning factual indicators of misinformation. This limitation provides an important motivation for examining model explanations and cross-dataset behaviour in the present research.

Propagation-based approaches investigate how information spreads through social networks, including temporal and structural characteristics of information cascades. Their advantage is that they introduce information that cannot be obtained from the article text alone, but they require social-network data and are therefore more complex and platform-dependent than content-only approaches.

Source-based approaches consider the historical credibility or characteristics of the publisher or source. Although source information can provide valuable signals, relying heavily on source identity can create another potential shortcut: a model might learn that articles from a particular source are usually labelled fake rather than learning characteristics of the article itself. This issue is closely related to the central concern of the present study: whether high predictive performance reflects generalisable learning or dataset-specific patterns.

2.4 Linguistic Characteristics of Fake News

The literature reviewed in the original project provides evidence that fake and legitimate news can exhibit different linguistic patterns. Zhou and Zafarani (2020) summarise research showing differences in informality, lexical diversity, subjectivity and emotional vocabulary between fake and legitimate news. These findings provide a theoretical justification for applying NLP techniques to fake news detection: if linguistic characteristics contain measurable differences between classes, they can be represented numerically and supplied to supervised learning algorithms.

However, the existence of linguistic differences does not necessarily establish that these differences will remain stable across datasets. Janicka, Pszona and Wawer (2019) provide direct empirical evidence for this concern. Training classifiers on one fake news dataset and testing them on others, they found accuracy could fall by more than twenty percentage points, and attributed the drop to models relying on dataset-specific lexical and stylistic cues rather than on stable indicators of misinformation. For example, if a particular benchmark dataset contains fake articles predominantly from websites that use sensational language, a classifier may learn sensational vocabulary as a strong indicator of the fake class. If a second dataset contains professionally written misinformation, that relationship may weaken or disappear.

Therefore, the key question is not simply *are there linguistic differences between fake and real news?* It is *which linguistic characteristics remain predictive when the model encounters a different data distribution?* This distinction represents an important transition from conventional benchmark classification towards the generalisation investigation proposed in this dissertation.

2.5 Traditional NLP and Machine Learning Approaches

The original research investigated TF-IDF as the primary feature representation. TF-IDF represents the importance of a term according to its frequency within a document and its relative frequency across the corpus (Salton and Buckley, 1988). The resulting representation converts each article into a high-dimensional numerical vector suitable for supervised machine learning.

The main advantage of TF-IDF is its simplicity, efficiency and interpretability. Individual vocabulary terms can be directly associated with model coefficients or feature importance values, making the representation suitable for investigating which words contribute to predictions.

This interpretability was one of the reasons why the original project selected traditional classifiers. Naive Bayes provides a probabilistic baseline based on the conditional distribution of features, while Logistic Regression provides directly interpretable feature coefficients. Random Forest provides a non-linear ensemble alternative. The original methodology therefore provided a controlled environment in which the classifiers could be compared under identical preprocessing and feature conditions.

However, there are limitations to this approach. TF-IDF primarily represents words according to their statistical importance and does not directly capture the deeper contextual relationships between words: two sentences containing similar vocabulary may communicate different meanings depending on word order and context. Traditional models can therefore be effective for benchmark classification while remaining limited in their ability to model contextual meaning.

More importantly for this research, the high dimensionality and dataset dependence of TF-IDF features may cause models to rely on vocabulary that is strongly associated with a particular dataset. In the large-scale benchmark reported by Dell'Oglio et al. (2026), Logistic Regression, SVM and Naive Bayes classifiers trained on one dataset and tested on another achieved median F1-scores close to 0.50 — little better than chance — despite performing competitively in-domain. This makes cross-dataset evaluation particularly valuable for determining whether the patterns learned by traditional classifiers remain effective outside the original training distribution.

2.6 Transformer-Based Approaches

The development of transformer architectures represents a significant change in NLP research. BERT introduced contextual language representations capable of modelling relationships between words based on their surrounding context (Devlin et al., 2019), while RoBERTa further optimised the pretraining procedure and demonstrated improved performance across a range of NLP tasks (Liu et al., 2019).

The original research identified BERT and RoBERTa as important future directions for benchmarking against the traditional classifiers.

Transformer-based approaches provide an important opportunity for the present research because they offer a fundamentally different representation from TF-IDF. Whereas TF-IDF relies heavily on term-level statistical representation, transformers produce contextual representations that can capture relationships between words and phrases. This creates a useful experimental comparison between a relatively transparent and computationally efficient TF-IDF baseline and a transformer-based model offering contextual language representation and greater modelling capacity.

However, greater modelling capacity does not automatically guarantee greater generalisation. Dell'Oglio et al. (2026) found that fine-tuned transformers such as BERT and DeBERTa (He et al., 2020) achieved the strongest in-domain results of any model family tested, with a median F1 of 0.86, yet suffered the same order of cross-dataset degradation as the traditional classifiers, falling to a median of 0.53. A transformer may therefore achieve excellent performance on a benchmark dataset while still learning correlations specific to the training distribution. Consequently, comparing transformer performance only within the original dataset would not answer the central research problem.

The present research therefore treats transformer performance as part of a broader question: does increased modelling complexity result in improved generalisation to unseen datasets, or does high benchmark performance remain sensitive to dataset-specific patterns?

2.7 Benchmark Datasets and Their Limitations

The quality and structure of datasets are fundamental to fake news detection research because supervised models learn directly from the examples and labels provided during training.

The ISOT Fake News Dataset used in the original research was constructed by Ahmed, Traore and Saad (2017; 2018) at the University of Victoria's Information Security and Object Technology research lab. It contains approximately 44,898 labelled articles, comprising real articles collected from Reuters.com and fake articles drawn from sources flagged as unreliable by fact-checking organisations. Its transparent construction and accessibility made it appropriate for reproducible experimentation, and it provided a relatively large labelled corpus with which multiple classifiers could be trained under identical conditions.

However, the dataset also has important limitations. The original research recognised that the data is predominantly English-language news from the 2016–2017 period, creating potential temporal and linguistic limitations. This means that a model trained and tested randomly within ISOT may benefit from characteristics shared across the dataset's sources and time period — including, potentially, superficial artefacts of how the two classes were collected rather than genuine markers of misinformation. Dell'Oglio et al. (2026) note explicitly that ISOT is one of the datasets in their benchmark most associated with strong, dataset-specific artefacts, with models trained on it generalising particularly poorly to other sources. Consequently, an apparently strong test result does not necessarily establish that the model would perform similarly on news from a different source, topic, period or dataset. This is one of the most important limitations motivating the present research, and is the specific reason WELFake (Verma et al., 2021) — a dataset built from four independently collected sources rather than a single paired real/fake collection — is proposed in Section 2.14 as the secondary evaluation dataset.

2.8 Dataset Bias and Generalisation

Zhou and Zafarani (2020) identify the need for more reliable and generalisable benchmark datasets and note that many existing datasets are limited by domain, political focus, scale or other characteristics. They also highlight explainability as an important challenge for fake news detection.

This observation creates an important methodological issue. A model achieving 95% F1 on Dataset A indicates that the model can distinguish the two classes effectively under the evaluation conditions represented by Dataset A. It does not necessarily follow that the model would achieve comparable performance on an unseen news distribution. The difference between these two statements is central to this dissertation, and is no longer only a theoretical possibility: Janicka, Pszona and Wawer (2019) documented accuracy drops exceeding twenty percentage points under cross-dataset conditions, and the more recent, larger-scale study by Dell'Oglio et al. (2026) documented a comparable median drop of over thirty F1 points across ten datasets and twelve model families, including traditional machine learning, deep learning and transformer approaches.

Cross-dataset evaluation provides a way of investigating this distinction. Instead of randomly dividing a single dataset into training and testing partitions, the model can be trained on one dataset and evaluated on another. This creates a more demanding evaluation condition because the model encounters a different distribution. The resulting difference in performance can be expressed as a generalisation gap:

Generalisation Gap = In-Domain Performance − Cross-Dataset Performance

A large gap provides evidence that benchmark performance does not fully represent performance under distribution shift. However, even a measured performance decrease does not explain its cause. This is why cross-dataset evaluation alone is insufficient for the present research, and why it is combined here with explainability and error analysis.

2.9 Explainability in Fake News Detection

Explainability is another major theme identified in the literature. Zhou and Zafarani (2020) highlight the importance of explainable detection systems capable of providing human-interpretable reasoning rather than functioning solely as opaque prediction systems.

Two techniques dominate the practical explainability literature for text classifiers. LIME (Local Interpretable Model-Agnostic Explanations) approximates the behaviour of any classifier locally around a specific prediction by fitting a simple, interpretable surrogate model to perturbed versions of the input, and was demonstrated by its authors specifically on text classification tasks (Ribeiro, Singh and Guestrin, 2016). SHAP (SHapley Additive exPlanations) instead assigns each feature an importance value derived from cooperative game theory, providing an alternative, theoretically grounded attribution that has been shown to generalise several earlier attribution methods, including LIME, as special cases (Lundberg and Lee, 2017). Both are model-agnostic, meaning they can be applied consistently to the traditional TF-IDF classifiers and to the transformer-based model used in this research, which is one reason they are adopted here (see Section 3.15).

Explainability is particularly important in fake news detection because incorrect predictions can have consequences beyond a conventional classification error. A false positive may incorrectly label legitimate journalism as fake, while a false negative may allow misleading information to be classified as legitimate. The original project therefore already recognised precision as an important metric and explicitly stated that automated detection should be treated as decision support rather than autonomous censorship.

Explainability can extend this principle by allowing users or researchers to inspect which characteristics influenced a model's prediction. However, explainability itself does not automatically prove that a model is correct: an explanation may show what the model used, but not necessarily whether that information represents genuine evidence of misinformation. This distinction is central to the current study. The research will therefore use explainability not merely as a user-interface feature, but as an investigative research instrument, examining whether the features influencing predictions remain consistent across datasets and whether cross-dataset failures reveal potential dataset-specific patterns.

2.10 From Feature Importance to Shortcut Learning

The original project proposed feature importance analysis using Logistic Regression coefficients and Random Forest feature importance scores to identify vocabulary associated with the fake and real classes. This provides a useful starting point for the revised study.

However, feature importance within a single dataset is insufficient to determine whether the identified features are genuinely generalisable. A word with a strong positive coefficient for the fake class in one dataset could represent a meaningful characteristic of fake news in general, or it could simply be disproportionately associated with fake articles in that particular dataset — a dataset-specific correlation rather than a generalisable signal.

This concern is formalised in the wider machine learning literature as *shortcut learning*: the tendency of a model to exploit decision rules that perform well on the distribution it was trained and tested on, but that fail to transfer to more demanding or differently distributed conditions, because the rule relied on a spurious correlation present in the training data rather than the intended, causally relevant signal (Geirhos et al., 2020). Dell'Oglio et al. (2026) invoke this concept directly when interpreting their own cross-dataset results, arguing that the sharp in-domain-to-cross-domain drop observed for fine-tuned transformers indicates that "generalist" models such as BERT and DeBERTa may overfit dataset-specific patterns rather than learning robust indicators of misinformation.

A cross-dataset experiment can help distinguish a shortcut from a genuine signal. If a feature remains predictive across datasets, this provides stronger evidence that it may represent a more generalisable pattern; if its importance disappears or changes substantially, this suggests that the feature may be dataset-dependent. This leads directly to the combination of cross-dataset evaluation, explainability and error analysis used in this research.

2.11 Cross-Dataset Generalisation: Empirical Evidence

The preceding sections have referred repeatedly to two studies that provide direct, empirical support for the research gap motivating this dissertation, and it is useful to consolidate what they show.

Janicka, Pszona and Wawer (2019) were among the first to systematically document cross-domain failure in fake news detection. Training traditional machine learning classifiers on one dataset and testing them on others drawn from different sources, they recorded accuracy drops exceeding twenty percentage points, and linked the failure to models exploiting dataset-specific lexical and stylistic artefacts rather than learning transferable indicators of misinformation.

Dell'Oglio et al. (2026) extended this line of enquiry considerably, evaluating twelve model families — spanning traditional machine learning, deep learning, fine-tuned transformers, specialised cross-domain architectures and large language models used in zero- and few-shot settings — across ten datasets, including ISOT itself, under four experimental paradigms: dataset-specific, cross-dataset, mixed-training and leave-one-dataset-out. Their central finding was that fine-tuned transformers, which achieved the strongest in-domain performance of any family tested (median F1 = 0.86), suffered the sharpest cross-dataset collapse (median F1 = 0.53), while traditional machine learning classifiers performed only marginally better than chance once evaluated outside their training distribution (median F1 around 0.50–0.52). They further reported that ISOT in particular was associated with strong, dataset-specific artefacts that limited the transferability of models trained on it, and that large language models used in a zero-shot setting were comparatively more stable across conditions, though still imperfect.

These two studies establish that the generalisation problem motivating this dissertation is real, substantial and consistent across different experimental designs and model families. What neither study provides is the fine-grained combination of feature-level explainability (via LIME and SHAP) and structured qualitative error analysis proposed here, applied specifically to the ISOT-to-WELFake transfer direction relevant to the original project. This is the specific space this dissertation occupies.

2.12 Limitations of Existing Research Approaches

The literature reviewed in this chapter identifies several recurring limitations.

Dataset limitations. Many datasets represent specific domains, time periods or sources. The literature identifies a need for more reliable and generalisable datasets that capture variation across domains, languages and time periods, and Dell'Oglio et al. (2026) show empirically that some individual datasets, including ISOT, carry particularly strong artefacts that limit transfer.

Dependence on textual characteristics. Content-based models are attractive because textual data is widely available and relatively easy to process. However, fake news can deliberately imitate credible journalism, meaning that stylistic characteristics may not consistently distinguish misinformation from legitimate reporting.

Limited contextual information. Shu et al. (2017) distinguish content features from social context, demonstrating that textual analysis represents only one component of the wider misinformation problem.

Explainability limitations. High-performing models may provide limited insight into why a particular prediction was produced, even when post-hoc methods such as LIME or SHAP are applied, since an explanation describes what a model used rather than proving that the information is causally meaningful.

Generalisation limitations. As established in Section 2.11, a model's performance on one benchmark does not necessarily establish its performance on another distribution, and the size of this gap can be large even for the most sophisticated model families currently in use.

Adversarial adaptation. Zhou and Zafarani (2020) also identify the possibility that misinformation creators may adapt their writing styles as detection systems improve, creating an ongoing relationship between detection and evasion.

These limitations indicate that classification accuracy alone provides an incomplete assessment of fake news detection systems.

2.13 Critical Comparison of Traditional and Transformer Approaches

The literature suggests a trade-off between model simplicity, interpretability and representational power.

Characteristic	TF-IDF + Traditional ML	Transformer-based models
Representation	Statistical word features	Contextual representations
Computational cost	Relatively low	Higher
Interpretability	Relatively high	More difficult without post-hoc tools
Contextual understanding	Limited	Stronger
In-domain performance (Dell'Oglio et al., 2026)	Median F1 ≈ 0.70–0.80	Median F1 up to 0.86
Cross-dataset performance (Dell'Oglio et al., 2026)	Median F1 ≈ 0.50–0.52	Median F1 ≈ 0.53

The important point, supported directly by the empirical results above, is that neither model family can be assumed to generalise simply because it is more sophisticated. A transformer may capture richer semantic relationships and consistently wins in-domain, but the evidence from Dell'Oglio et al. (2026) shows both families collapsing to a broadly similar, weak level of cross-dataset performance. Therefore, the research should not assume in advance that the transformer will be the best-generalising model, even though it is expected to be the strongest in-domain performer. Instead, the experimental design allows the data to determine which model generalises most effectively, and what evidence explains the difference — a more defensible research position than designing the study around the expectation that a particular model will win on every criterion.

2.14 Synthesis of the Literature

The literature reviewed in this chapter leads to several conclusions.

First, fake news detection can be approached through multiple information sources, including content, style, knowledge, propagation and source information. Shu et al. (2017) and Zhou and Zafarani (2020) establish that textual analysis is only one dimension of the broader problem.

Second, textual characteristics contain useful predictive information, but the differences that distinguish fake and legitimate news within one dataset are not guaranteed to hold in another, as Janicka, Pszona and Wawer (2019) demonstrated directly.

Third, traditional machine learning using representations such as TF-IDF provides a strong and interpretable in-domain baseline, but Dell'Oglio et al. (2026) show this baseline degrades to near-chance performance under cross-dataset conditions.

Fourth, transformer architectures such as BERT and RoBERTa provide more sophisticated contextual representations and the strongest in-domain results currently reported in the literature, but they are not immune to the same generalisation collapse, and in some benchmarks show an even sharper relative decline than simpler models.

Fifth, dataset characteristics represent a significant concern. The limitations of the ISOT dataset, particularly its temporal and linguistic scope and its association with strong, source-specific artefacts, demonstrate why performance within a single benchmark should not automatically be interpreted as evidence of real-world robustness. This is the direct motivation for pairing ISOT with WELFake, an independently constructed, multi-source dataset, in the present study.

Finally, explainability (via LIME and SHAP) and the shortcut-learning framework (Geirhos et al., 2020) provide a vocabulary and a set of tools for investigating *why* a generalisation gap occurs, rather than only measuring *that* it occurs.

These observations lead to the central research problem of this dissertation: a model may achieve strong benchmark performance without demonstrating robust generalisation, and conventional performance metrics alone cannot establish whether its predictions are based on generalisable characteristics or dataset-specific patterns.

2.15 Research Gap

Based on the literature reviewed, the research gap addressed by this study can be stated as follows:

Existing research provides strong empirical evidence that fake news detection models — across traditional machine learning, deep learning and transformer families alike — suffer substantial performance degradation under cross-dataset evaluation (Janicka, Pszona and Wawer, 2019; Dell'Oglio et al., 2026). However, this evidence has not yet been connected, at a fine-grained level, to feature-level explainability and structured error analysis for the specific ISOT-to-WELFake transfer scenario relevant to models built in the TF-IDF and RoBERTa traditions established by the original project.

The research therefore does not claim that cross-dataset evaluation or explainable AI has never been studied — the opposite is demonstrably true. Instead, the contribution lies in examining cross-dataset generalisation, explainability and error analysis together, within a controlled experimental framework built directly on the original project's dataset and model choices, so that the findings connect back to and extend that earlier work specifically.

The proposed research will compare model performance under in-domain and cross-dataset evaluation, and then investigate the observed differences using explainability and error analysis. This allows the research to move beyond the question "which model achieves the highest accuracy?" towards the more important question: "does the model continue to behave reliably when the data changes, and what can its explanations and errors tell us about the reasons for its success or failure?"

2.16 Conceptual Framework for the Research

The literature review supports the following conceptual relationship:

Dataset characteristics → Textual representation → Machine learning model → In-domain performance → Cross-dataset performance → Generalisation gap → Explainability analysis (LIME / SHAP) → Error analysis → Identification of potential dataset-specific patterns / shortcut learning

This framework connects the literature directly to the experimental methodology. The dataset represents the environmental conditions under which the model learns. The representation determines how textual information is encoded. The model transforms those representations into predictions. In-domain evaluation measures performance under familiar conditions. Cross-dataset evaluation tests whether performance remains stable under changed conditions, consistent with the substantial gaps documented by Janicka, Pszona and Wawer (2019) and Dell'Oglio et al. (2026). Explainability examines the factors contributing to individual predictions. Error analysis investigates where and how predictions fail, informed by the shortcut-learning framework of Geirhos et al. (2020). Together, these components provide a more complete assessment of model behaviour than benchmark accuracy alone.

2.17 Chapter Summary

This chapter critically reviewed the literature underpinning automated fake news detection and the revised research direction.

The literature establishes that fake news detection can incorporate content, style, knowledge, propagation and source information. Text-based approaches remain valuable because linguistic characteristics can provide measurable signals for classification, but they also have important limitations because fake news can imitate credible journalism and because linguistic patterns may vary between datasets — a concern given direct empirical weight by Janicka, Pszona and Wawer (2019) and by the large-scale benchmark of Dell'Oglio et al. (2026), both of which documented substantial cross-dataset performance collapse across multiple model families, including the ISOT dataset used as this project's primary benchmark.

Traditional machine learning and TF-IDF (Salton and Buckley, 1988) provide interpretable and computationally efficient in-domain baselines, while transformer architectures such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019) provide the strongest in-domain results but are not immune to the same generalisation collapse. The review also identified dataset characteristics as a significant concern, motivating the pairing of ISOT with the independently constructed WELFake dataset (Verma et al., 2021) for cross-dataset evaluation.

Explainability, via LIME (Ribeiro, Singh and Guestrin, 2016) and SHAP (Lundberg and Lee, 2017), and the shortcut-learning framework (Geirhos et al., 2020) provide the conceptual and methodological tools needed to investigate *why* a generalisation gap occurs rather than only measuring *that* it occurs.

The literature therefore supports the central direction of this research: investigating cross-dataset generalisation and model decision behaviour rather than simply comparing classification accuracy. The next chapter will translate this research gap into a detailed methodology, including dataset selection, preprocessing, model development, cross-dataset experimental design, evaluation metrics, explainability methods and error analysis.

CHAPTER 3: METHODOLOGY AND RESEARCH DESIGN

3.1 Introduction

This chapter presents the methodology and research design used to investigate the cross-dataset generalisation and decision behaviour of machine learning models for fake news detection. The methodology builds upon the experimental foundation established in the previous stage of the project while extending the evaluation beyond a single benchmark dataset.

The original study was designed as a quantitative computational experiment using the ISOT Fake News Dataset (Ahmed, Traore and Saad, 2017; 2018), NLP preprocessing, TF-IDF feature representation (Salton and Buckley, 1988) and three supervised machine learning classifiers: Naive Bayes, Logistic Regression and Random Forest. It used an 80% training, 10% validation and 10% testing split and evaluated the models using accuracy, precision, recall and F1 score.

The revised methodology retains this baseline so that the new experiments remain connected to the original research, while extending it through cross-dataset evaluation, explainability and systematic error analysis. This allows the investigation to move beyond the question of which model performs best on the original benchmark and instead examine whether model performance and decision behaviour remain stable when the data distribution changes.

3.2 Research Philosophy

This research adopts a positivist research philosophy, selected because the research investigates measurable properties of news articles and evaluates machine learning models using objective numerical measurements, consistent with the original project's positioning.

The research does not attempt to determine the philosophical or political meaning of fake news through subjective interpretation. Instead, it measures observable computational outcomes, including classification performance, changes in performance between datasets, feature contributions and prediction errors. One partial exception is noted in Section 3.16: the qualitative categorisation of error patterns necessarily involves a degree of researcher judgement, which is acknowledged as a limitation of an otherwise positivist design rather than treated as if it were purely objective measurement.

3.3 Research Approach

A deductive approach is adopted. The research begins with observations and limitations identified in the existing literature — particularly the substantial cross-dataset performance drops documented by Janicka, Pszona and Wawer (2019) and Dell'Oglio et al. (2026), and the shortcut-learning framework of Geirhos et al. (2020) — which lead to the research questions and experimental propositions developed in Chapter 1. The experimental process is then used to test these propositions: a model is trained and evaluated under controlled in-domain conditions, the same model is evaluated on an unseen dataset, performance changes are measured, and explainability and error analysis investigate possible reasons for those changes. This deductive structure is consistent with the original project.

3.4 Research Design

The research uses a quantitative experimental computational design, retaining the controlled comparison of supervised classifiers used in the original study but adding cross-dataset evaluation as a further dimension. The experimental design consists of four main stages: baseline development and in-domain evaluation; cross-dataset evaluation; explainability analysis; and systematic error analysis. The baseline experiment establishes model performance on the original dataset. The cross-dataset experiment then evaluates whether the trained models maintain their performance when presented with data from a different distribution. The explainability stage examines which features contribute to predictions, while the error-analysis stage investigates cases where the models make incorrect predictions. This design allows the research to evaluate both what the models predict and how their behaviour changes when the data changes.

3.5 Research Datasets

3.5.1 Primary Dataset: ISOT Fake News Dataset

The primary dataset used in this research is the ISOT Fake News Dataset, developed by Ahmed, Traore and Saad (2017; 2018) at the Information Security and Object Technology Research Lab, University of Victoria.

The dataset contains approximately 44,898 labelled news articles, consisting of 21,417 real articles and 23,481 fake articles. The real articles were collected from Reuters.com, while the fake articles originated from sources identified as unreliable. The dataset contains article titles, article text, subject information and publication dates. The original methodology merged the separate Fake.csv and True.csv files and assigned binary labels, with real articles represented as class 0 and fake articles represented as class 1.

ISOT is retained as the primary dataset for three reasons: it provides a relatively large labelled corpus suitable for supervised learning; it provides a consistent benchmark that allows the revised research to remain directly connected to the original project; and using the same primary dataset allows the current research to establish a baseline before testing generalisation on additional data.

However, the dataset is predominantly associated with English-language news from the 2016–2017 period, and Dell'Oglio et al. (2026) specifically identify ISOT as a dataset associated with strong, source-specific artefacts that limit the cross-dataset transfer of models trained on it. This is one reason cross-dataset evaluation is included in the revised research.

3.5.2 Secondary Dataset for Cross-Dataset Evaluation: WELFake

The secondary dataset used for cross-dataset evaluation is **WELFake** (Word Embedding over Linguistic Features for Fake News Detection; Verma et al., 2021). WELFake contains 72,134 labelled news articles (35,028 real, 37,106 fake), constructed by merging four independently collected source datasets — Kaggle, McIntire, Reuters and BuzzFeed Political — specifically to reduce the single-source bias and overfitting risk present in datasets built from only one real/fake pairing.

WELFake was selected against the six criteria set out in the original proposal:

1. **Labelled real/fake examples** — satisfied; WELFake provides a binary Real/Fake label for every record.
2. **Sufficient textual information** — satisfied; the dataset provides both title and full article text.
3. **Label compatibility** — satisfied without remapping, since both ISOT and WELFake use a direct binary real/fake distinction at the article level. This is a key reason WELFake was preferred over LIAR (Wang, 2017), whose six-class, short-statement format would require both label remapping and a genre change.
4. **Public accessibility** — satisfied; WELFake is openly available via Zenodo and Kaggle under a research-use licence.
5. **Reproducible structure** — satisfied; the dataset is distributed as a single CSV with a consistent schema.
6. **Meaningful distributional difference from ISOT** — satisfied; WELFake's four constituent sources only partially overlap with ISOT's Reuters/unreliable-source pairing, and its combination of multiple original collections is specifically intended to reduce the kind of single-source artefact that Dell'Oglio et al. (2026) identify in ISOT.

This selection is treated as the working plan for the study, consistent with Objective 3 and the scope set out in Section 1.9. It remains subject to revision if the dataset inspection described in Section 3.6 reveals structural incompatibilities (for example, in text length distribution or label reliability) that cannot be reasonably addressed through preprocessing; a documented note by Zeng et al. (2022), for instance, has previously flagged a labelling inversion in earlier WELFake distributions, so label integrity will be verified explicitly as part of dataset inspection before training.

3.6 Dataset Preparation

The primary ISOT dataset is provided through separate real and fake news files. These files are combined into a single dataset and assigned binary class labels: 0 = Real, 1 = Fake. The same conceptual labelling structure will be applied to WELFake after its native Real/Fake column has been checked against the verification note above.

Dataset inspection will be performed before model training for both datasets, and will include examination of: number of records; class distribution; missing values; duplicate articles; empty text fields; text length; available metadata; publication dates where available; and source information where available. This inspection is necessary because differences between datasets may themselves influence the cross-dataset results, and because it provides the evidence needed to confirm or revise the dataset selection made in Section 3.5.2.

3.7 Data Cleaning and Preprocessing

Text preprocessing converts raw news articles into a consistent representation suitable for machine learning. The original research used a five-stage preprocessing pipeline consisting of text cleaning, lowercasing, tokenisation, stopword removal and TF-IDF vectorisation. The revised study retains these core preprocessing principles for the traditional machine learning baseline.

Text cleaning. Punctuation, special characters, HTML tags and other non-alphabetic elements will be removed where appropriate, to reduce unnecessary noise in the textual representation.

Lowercasing. All text will be converted to lowercase so that different capitalisation forms of the same word (e.g. "President", "president", "PRESIDENT") are treated consistently, preventing capitalisation differences from unnecessarily increasing the vocabulary size.

Tokenisation. The cleaned text will be divided into individual tokens, providing the basic units required for word-level statistical representation.

Stopword removal. Common function words such as "the", "is", "at" and "on" may be removed, since they generally provide limited discriminative information for the classification task, while also reducing the dimensionality of the resulting feature space.

Handling missing and invalid records. Articles with missing or empty text fields will be identified before training. Duplicate records will also be investigated to reduce the possibility of information leakage between training and testing data.

The same preprocessing logic will be applied consistently to the training and evaluation data.

For the RoBERTa-based model (Section 3.10.4), preprocessing will instead follow standard subword tokenisation via the model's own tokenizer, with minimal manual cleaning (retaining case and punctuation, both of which carry information for a contextual model), consistent with the way BERT-family models are used in comparable benchmark work (Dell'Oglio et al., 2026).

3.8 Prevention of Data Leakage

Data leakage is an important consideration in this research because the objective is to evaluate genuine generalisation. For the traditional machine learning pipeline, the TF-IDF vocabulary and IDF statistics will be fitted only on the training data; the validation and test datasets will then be transformed using the already-fitted representation. The same principle will apply to cross-dataset evaluation: WELFake will not be used to fit the primary model's vocabulary, tokenizer, or model parameters at any stage. This ensures that information from the unseen dataset does not influence the training process, so that cross-dataset performance represents the model's ability to generalise rather than its ability to exploit information from the evaluation data. Near-duplicate checking (via text hashing) will also be applied between ISOT and WELFake prior to evaluation, since both datasets partially draw on Reuters as a source of real news, to confirm that any shared articles are excluded from the cross-dataset test set.

3.9 Feature Representation

3.9.1 TF-IDF Representation

TF-IDF will be used as the primary feature representation for the traditional machine learning baseline. TF-IDF assigns numerical weights to terms according to their frequency within a document and their relative frequency across the corpus (Salton and Buckley, 1988). The basic formulation used in the original project is:

$$\text{TF-IDF}(t,d,D) = \text{TF}(t,d) \times \log(N / \text{df}(t))$$

where t represents a term, d represents a document, D represents the document collection, N represents the number of documents, and $df(t)$ represents the number of documents containing term t. TF-IDF is particularly appropriate for establishing an interpretable baseline because individual vocabulary features can be related directly to model predictions.

3.9.2 Contextual Representation

The revised research will also investigate a transformer-based approach as a modern comparison with the traditional TF-IDF pipeline (see Section 3.10.4). The transformer model will use contextual representations rather than relying exclusively on independent word-level statistical weights, producing a methodological comparison between TF-IDF → Machine Learning and Text → Transformer → Classification. The purpose is not to assume that the transformer will outperform the traditional models, particularly under cross-dataset conditions, where the literature reviewed in Section 2.13 suggests the advantage may narrow substantially; rather, the experiment will determine whether increased representational capacity produces improved generalisation.

3.10 Machine Learning Models

The original study selected three supervised classifiers, retained here as the traditional baseline, plus one transformer-based model added for this revised study.

3.10.1 Multinomial Naive Bayes

Multinomial Naive Bayes provides the probabilistic baseline. It is computationally efficient and suitable for sparse textual feature representations. The model assumes conditional independence between features given the class, which is a simplifying assumption but allows efficient classification of high-dimensional text data.

3.10.2 Logistic Regression

Logistic Regression is used as an interpretable linear classifier. Its learned coefficients can be examined to identify vocabulary features associated with the predicted class, making it particularly useful for connecting classification performance with feature-level analysis.

3.10.3 Random Forest

Random Forest provides a non-linear ensemble comparison. The model constructs multiple decision trees using bootstrap samples and randomly selected feature subsets, with predictions generated through aggregation across the trees. Its inclusion allows the research to determine whether a non-linear ensemble approach provides improved performance over simpler linear and probabilistic approaches, and whether that advantage (if any) persists under cross-dataset conditions.

3.10.4 Transformer-Based Model: RoBERTa-base

The transformer-based model used in this research is **RoBERTa-base** (Liu et al., 2019), continuing the direction identified as future work in the original project, which explicitly named BERT and RoBERTa as candidates for benchmarking against the traditional classifiers. RoBERTa was selected over base BERT because it is trained with an optimised pretraining procedure removing the next-sentence-prediction objective and using larger batches and more data, which has been shown to produce consistent improvements over BERT on downstream classification tasks (Liu et al., 2019). The model will be fine-tuned using the Hugging Face `transformers` implementation (`roberta-base`), with a classification head added for binary fake/real prediction. This choice is treated as the working default for the study; if computational constraints identified during implementation make full fine-tuning impractical, a smaller distilled variant will be substituted and the substitution documented explicitly in Chapter 4, rather than left unresolved as in the original proposal.

3.11 Dataset Splitting

For the primary dataset, the original experimental design used a stratified 80% training, 10% validation and 10% test split, with the test set held out until final evaluation. The revised study will maintain this structure for the primary baseline where computationally appropriate. Stratification will be used to maintain the relative distribution of real and fake classes across the partitions. The test set will remain isolated from model development decisions, and the validation set will be used for model configuration and hyperparameter selection, following the same tuning-budget principle used by Dell'Oglio et al. (2026): each model is tuned once per dataset on the same train/validation split, so that no model receives a systematically larger tuning budget than another.

For WELFake, since it is used exclusively as an unseen cross-dataset test set rather than for training (Section 3.8), the full WELFake dataset — after inspection and cleaning — will be used for cross-dataset evaluation rather than being split into further partitions.

3.12 Baseline In-Domain Experiment

The first experiment establishes the baseline performance of each selected model. ISOT is preprocessed, split into training, validation and test partitions, transformed into the appropriate feature representation, and used to train each traditional classifier and the RoBERTa-based model under identical data conditions; each model is then evaluated on the same held-out ISOT test partition. The purpose of this experiment is to establish the performance that can be achieved when training and testing occur within the same dataset distribution.

3.13 Cross-Dataset Experiment

The second experiment is the central experiment of the revised research. The selected models, trained using the primary ISOT dataset, will be evaluated directly against the full, unseen WELFake dataset, with no further training or fine-tuning on WELFake. This experiment directly addresses Research Question 1 by examining whether the models retain their predictive performance when the evaluation distribution differs from the training distribution. The results will be compared with the in-domain results from Section 3.12, and the difference between the two conditions will be used to quantify the generalisation gap (Section 3.14).

3.14 Generalisation Analysis

Generalisation will be analysed by comparing the performance of each model under two conditions: Condition A (in-domain — training and testing on ISOT) and Condition B (cross-dataset — training on ISOT, testing on WELFake). For each model, the change in performance will be calculated, for example:

F1 Generalisation Gap = F1(In-Domain) − F1(Cross-Dataset)

The same principle will be applied to accuracy, precision and recall. A smaller performance reduction will indicate stronger observed generalisation under the experimental conditions.

Raw metric differences alone cannot establish whether an observed gap reflects a genuine, systematic effect rather than sampling variation, particularly given that the ISOT and WELFake test sets differ in size. To address this, statistical significance will be assessed in two ways. First, **McNemar's test** will be applied to the paired correct/incorrect prediction outcomes of each model under in-domain versus cross-dataset conditions, to test whether the change in error rate is statistically significant rather than attributable to chance. Second, where multiple models are compared against one another within the same condition, a **Friedman test** followed by a post-hoc **Nemenyi test** will be used, following the approach adopted in comparable recent benchmarking work (Dell'Oglio et al., 2026), since this combination is appropriate for comparing more than two classifiers across evaluation conditions without assuming normally distributed performance differences. For the RoBERTa-based model, which involves stochastic training, five training runs with different random seeds will be conducted, and mean F1 with standard deviation will be reported alongside the single-run result used for the main comparison, in order to distinguish genuine generalisation differences from run-to-run variance.

The results will not be interpreted solely through numerical or statistical differences; explainability and error analysis will subsequently be used to investigate potential reasons for the observed changes.

3.15 Explainability Analysis

Explainability will be used as an analytical component of the research rather than simply as a visual feature of the final application. Two complementary, model-agnostic techniques will be applied consistently across all models:

- **LIME** (Ribeiro, Singh and Guestrin, 2016), which approximates each individual prediction locally with an interpretable surrogate model, will be used as the primary method, since it operates directly on text and is well suited to producing per-prediction word-level explanations for both the TF-IDF classifiers and the fine-tuned RoBERTa model.
- **SHAP** (Lundberg and Lee, 2017) will be used as a secondary, cross-checking method, applied via its linear/tree explainers for the traditional models and its gradient-based explainer for RoBERTa, to verify whether the two methods converge on the same influential features. Convergence between the two methods will be treated as stronger evidence that an identified feature is genuinely influential rather than an artefact of one particular attribution technique.

For the traditional models, Logistic Regression coefficients and Random Forest feature importance scores (as already proposed in the original project) will additionally be examined as a third, model-native point of comparison against LIME and SHAP.

The revised research extends this analysis by comparing influential features across the in-domain (ISOT) and cross-dataset (WELFake) conditions, investigating: which features contribute strongly to predictions; whether the same features remain important across datasets; whether incorrect predictions are associated with particular linguistic patterns; and whether explanations provide evidence of dataset-specific behaviour consistent with shortcut learning (Geirhos et al., 2020). All explainability configuration (sample size for explanation generation, random seed, and library versions) will be documented explicitly so that the analysis remains reproducible.

3.16 Error Analysis

Performance metrics provide aggregate information but do not explain individual failures. The research will therefore conduct systematic error analysis, using the four standard outcome categories: false positive (a real news article predicted as fake); false negative (a fake news article predicted as real); true positive; and true negative. Particular attention will be given to false positives and false negatives.

Sampling. Where the number of errors for a given model and condition is small (fewer than approximately 100), all errors will be examined. Where the number of errors is large, a stratified random sample of 100 false positives and 100 false negatives will be drawn (or the full set, if fewer are available) for each model and condition, using a fixed random seed for reproducibility.

Categorisation procedure. The candidate error patterns identified in the original proposal — unusual writing style, political or topic-specific vocabulary, short article length, highly emotional language, source-specific terminology, differences in article structure, and vocabulary common in one dataset but rare in the other — will be used as an initial coding framework, applied through open coding by the researcher against the sampled errors, informed directly by the LIME/SHAP explanations generated in Section 3.15 for the same predictions. Because this stage involves a single coder, it is acknowledged as a qualitative judgement rather than a fully objective measurement (see Section 3.2), and a randomly selected 20% subsample of coded errors will be re-checked by the researcher after a delay to assess internal consistency of the coding.

The purpose of this analysis is not to manually change the labels simply because the model made an error, but to investigate whether systematic characteristics can explain groups of errors, directly addressing Research Question 3.

3.17 Evaluation Metrics

The models will be evaluated using standard classification metrics.

Accuracy measures the proportion of correctly classified examples: Accuracy = (TP + TN) / (TP + TN + FP + FN).

Precision measures how many articles predicted as fake are actually fake: Precision = TP / (TP + FP). Precision is particularly important because a low precision value indicates that a large proportion of articles labelled as fake are actually legitimate; the original research therefore identified precision as an important metric because false-positive predictions can have consequences for legitimate journalism.

Recall measures how many actual fake articles are correctly identified: Recall = TP / (TP + FN).

F1 score combines precision and recall through their harmonic mean: F1 = 2 × (Precision × Recall) / (Precision + Recall), providing a balanced measure when both false positives and false negatives are important.

These four metrics were already established in the original project and will remain central to the revised evaluation, supplemented by ROC-AUC where appropriate and by the significance-testing procedures set out in Section 3.14.

3.18 Confusion Matrix Analysis

Confusion matrices will be generated for the main experimental conditions, showing true positives, true negatives, false positives and false negatives. This allows the research to determine whether performance changes between datasets are driven primarily by increases in false positives, false negatives or both. For example, a model may retain relatively high accuracy while experiencing a substantial increase in false negatives on the secondary dataset; such a result would be important because aggregate accuracy alone could hide a meaningful change in model behaviour, particularly given that WELFake's class balance (35,028 real / 37,106 fake, a near-even split) differs only slightly from ISOT's, meaning any large shift in confusion-matrix composition would need to be explained by dataset shift rather than class-imbalance artefacts.

3.19 Model Comparison

The models will not be ranked using accuracy alone. The comparison will consider accuracy, precision, recall, F1 score, confusion matrix, in-domain performance, cross-dataset performance, the generalisation gap and its statistical significance (Section 3.14), explainability results and error patterns. This produces a multidimensional evaluation of model performance. The research will therefore avoid concluding that a model is "best" simply because it obtains the highest benchmark accuracy; instead, the final assessment will consider performance, generalisation and interpretability together.

3.20 Statistical and Experimental Reliability

The quantitative results will be reported consistently across the experimental conditions. As set out in Section 3.14, the RoBERTa-based model will be trained across five random seeds, with mean and standard deviation reported; the traditional classifiers, which are deterministic given a fixed train/test split and hyperparameter configuration, will use a single fixed random seed (documented in Chapter 4) for the split itself. The primary test set will remain isolated from model development, and all preprocessing and model configuration decisions will be documented. For the final experiments, the exact random seeds, dataset versions, preprocessing configuration, model parameters and software environment will be recorded, in order to improve reproducibility and reduce the possibility that reported performance results are the consequence of an accidental experimental configuration.

3.21 Reproducibility

Reproducibility is an important requirement of the experimental design. The implementation will be developed using Python and appropriate open-source machine learning and NLP libraries. The original project used Python 3, Pandas, NLTK and Scikit-learn; the revised implementation will maintain this principle wherever possible and will additionally use the Hugging Face `transformers` and `datasets` libraries for RoBERTa, and the `lime` and `shap` libraries for explainability.

The following information will be documented: dataset source and version; dataset preprocessing steps; train/validation/test split; random seed(s); feature representation; model architecture; hyperparameters; training configuration; evaluation metrics; explainability configuration; software library versions; Python version; and hardware environment where relevant. The final project artefact and supporting code will be organised so that the experimental workflow can be reproduced.

3.22 Artefact Development

The practical artefact will be an interactive fake news detection web application based on the research pipeline. The system will allow a user to enter or submit news text and receive a model prediction, a confidence score, and a LIME-based visual explanation highlighting the words that most influenced the prediction. The proposed workflow is: user input → text preprocessing → feature/contextual representation → trained model → prediction and confidence score → LIME explanation displayed to the user.

The artefact will be designed as a research demonstration rather than a fully autonomous fact-checking authority. The interface will communicate that the model provides a prediction based on patterns learned from labelled datasets, including an explicit note that the model's training data (ISOT) is known, from the cross-dataset results reported in Chapter 4, to generalise imperfectly to news from other sources. This distinction is important because a machine learning prediction does not independently establish whether a real-world claim is factually true. The artefact therefore demonstrates the practical application of the research while maintaining appropriate limitations.

3.23 Ethical Considerations

The research primarily uses publicly available, pre-labelled textual datasets (ISOT and WELFake) and does not involve direct recruitment of human participants. The original research therefore identified the project as low risk from the perspective of informed consent and personally identifiable information, and this assessment is maintained.

Nevertheless, ethical considerations remain relevant. The most significant concern is the potential misuse of automated fake news classification: a false positive could result in legitimate journalism being incorrectly labelled as fake, while a false negative could allow misleading information to receive a classification indicating that it resembles legitimate news. For this reason, the system will not be presented as an autonomous censorship or fact-checking mechanism, but as a decision-support research artefact that can assist users in examining potentially misleading content. The limitations of the datasets, including temporal, linguistic and source-related biases and the cross-dataset findings reported in Chapter 4, will also be explicitly acknowledged in the artefact's interface and in the dissertation's conclusions.

This research will be submitted for ethical approval in accordance with the institution's research ethics policy and procedures prior to any data processing beyond the publicly available datasets described above; the specific approval reference will be recorded here once granted.

3.24 Data Protection and Privacy

The research does not involve the collection of personal information from human participants. The primary datasets consist of publicly available news content rather than participant responses, and no personal identifiers will intentionally be collected as part of the experimental process. Where datasets contain source or publication metadata, this information will only be retained where it is relevant to the research and permitted under the dataset's terms of use. The project will follow the institution's research requirements concerning responsible data handling.

3.25 Methodological Limitations

Several methodological limitations must be recognised.

First, the research is based primarily on labelled datasets and therefore inherits limitations present in their construction and labelling procedures, including the previously documented labelling concerns for earlier WELFake distributions noted in Section 3.5.2, which will be explicitly re-verified before use.

Second, the primary dataset is predominantly English-language and represents a specific historical period; consequently, findings may not generalise to all languages, countries, topics or contemporary news environments, including news shaped by large language model-generated misinformation, which lies outside the scope of both ISOT and WELFake.

Third, cross-dataset evaluation provides a stronger test of generalisation than random within-dataset testing, but it still represents only the specific ISOT-to-WELFake transfer direction selected for this experiment; the wider benchmark reported by Dell'Oglio et al. (2026) suggests that different dataset pairs can produce meaningfully different degrees of generalisation gap, so the results here should not be over-generalised to all possible dataset transfers.

Fourth, explainability methods provide information about model behaviour but do not prove that the identified features represent causal evidence of misinformation; LIME and SHAP can also disagree with one another on a given prediction, and such disagreements will themselves be reported as evidence of attribution uncertainty rather than resolved arbitrarily in favour of one method.

Fifth, the qualitative error-categorisation process described in Section 3.16 relies on a single researcher-coder, which is a limitation for an otherwise positivist, quantitative design.

Finally, the binary fake/real classification formulation simplifies a more complex real-world problem. Some news may contain partial truths, misleading framing, outdated information or disputed claims that cannot be accurately represented by a binary label.

These limitations will be considered when interpreting the final results.

3.26 Methodological Alignment with Research Questions

Research Question	Experimental Method
RQ1: How does model performance change across datasets?	In-domain vs cross-dataset evaluation (ISOT → WELFake), with McNemar/Friedman–Nemenyi significance testing
RQ2: What features influence predictions?	LIME and SHAP feature-level analysis, compared with model-native coefficients/importances
RQ3: Can explainability and error analysis identify potential shortcut/dataset-specific behaviour?	Explanation comparison across conditions + structured false-positive/false-negative analysis

This alignment ensures that every major experiment contributes directly to answering a research question, and that the methodology avoids adding experiments that do not contribute to the research argument.

3.27 End-to-End Research Pipeline

The complete experimental pipeline extends the original research pipeline (data collection, preprocessing, TF-IDF, model training, evaluation) as follows:

1. **Dataset acquisition** — ISOT and WELFake.
2. **Dataset inspection** — class distribution, missing values, duplicates, text characteristics, label-integrity verification

(WELFake).
3. **Preprocessing** — cleaning, lowercasing, tokenisation, stopword handling (traditional models); subword tokenisation (RoBERTa).
4. **Feature representation** — TF-IDF (traditional models) / contextual embeddings (RoBERTa).
5. **Model development** — Naive Bayes, Logistic Regression, Random Forest, RoBERTa-base.
6. **In-domain evaluation** — ISOT training set → held-out ISOT test set.
7. **Cross-dataset evaluation** — ISOT-trained models → full WELFake set.
8. **Generalisation analysis** — performance difference between conditions, with significance testing.
9. **Explainability** — LIME and SHAP feature attribution across both conditions.
10. **Error analysis** — sampled false positives/negatives, structured qualitative categorisation.
11. **Integrated analysis** — performance + generalisation + explanations + errors considered jointly.
12. **Artefact** — interactive fake news detection web application with LIME-based explanation display.

3.28 Chapter Summary

This chapter established the methodology for investigating fake news detection model generalisation and decision behaviour. The research adopts a positivist philosophy and deductive approach within a quantitative experimental computational design. The ISOT Fake News Dataset is retained as the primary benchmark to maintain continuity with the original project, and WELFake (Verma et al., 2021) is introduced as the secondary, unseen dataset for cross-dataset evaluation, selected and justified against the six criteria set out in the original proposal.

Traditional TF-IDF-based classifiers are retained as baseline models, while RoBERTa-base (Liu et al., 2019) provides a modern contextual comparison, continuing the direction the original project identified as future work. The experimental design separates in-domain evaluation from cross-dataset evaluation so that the difference in performance can be quantified and tested for statistical significance using McNemar’s test and the Friedman–Nemenyi procedure. Explainability analysis, using LIME (Ribeiro, Singh and Guestrin, 2016) as the primary method and SHAP (Lundberg and Lee, 2017) as a cross-check, and systematic, sampled error analysis are then used to investigate potential reasons for changes in model behaviour, informed by the shortcut-learning framework of Geirhos et al. (2020).

Accuracy, precision, recall and F1 score provide the primary quantitative evaluation measures, while confusion matrices, generalisation gaps, feature explanations and error patterns provide additional evidence. The final artefact will implement the research approach as an interactive fake news detection system while clearly presenting the system as a decision-support tool rather than an autonomous source of factual truth.

Overall, the methodology transforms the original single-dataset classifier comparison into a broader, more concretely specified experimental investigation of generalisation, model behaviour and explainability under dataset shift.

REFERENCES

Ahmed, H., Traore, I. and Saad, S. (2017) ‘Detection of online fake news using n-gram analysis and machine learning techniques’, in *Intelligent, Secure, and Dependable Systems in Distributed and Cloud Environments: First International Conference, ISDDC 2017*. Vancouver, BC, Canada, 26–28 October. Cham: Springer, pp. 127–138.

Ahmed, H., Traore, I. and Saad, S. (2018) ‘Detecting opinion spams and fake news using text classification’, *Security and Privacy*, 1(1), p. e9.

Dell’Oglio, P., Bondielli, A., Marcelloni, F. and Passaro, L.C. (2026) ‘An experimental comparison of the most popular approaches to fake news detection’, *Information Sciences* (preprint). Available at: https://arxiv.org/abs/2603.25501 (Accessed: 18 August 2026).

Devlin, J., Chang, M.W., Lee, K. and Toutanova, K. (2019) ‘BERT: Pre-training of deep bidirectional transformers for language understanding’, in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Minneapolis, MN, June. Association for Computational Linguistics, pp. 4171–4186.

Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M. and Wichmann, F.A. (2020) ‘Shortcut learning in deep neural networks’, *Nature Machine Intelligence*, 2(11), pp. 665–673.

He, P., Liu, X., Gao, J. and Chen, W. (2020) ‘DeBERTa: Decoding-enhanced BERT with disentangled attention’, *arXiv preprint* arXiv:2006.03654.

Janicka, M., Pszona, M. and Wawer, A. (2019) ‘Cross-domain failures of fake news detection’, *Computación y Sistemas*, 23(3), pp. 1089–1097.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L. and Stoyanov, V. (2019) ‘RoBERTa: A robustly optimized BERT pretraining approach’, *arXiv preprint* arXiv:1907.11692.

Lundberg, S.M. and Lee, S.I. (2017) ‘A unified approach to interpreting model predictions’, in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*. Long Beach, CA, December.

Ribeiro, M.T., Singh, S. and Guestrin, C. (2016) ‘“Why should I trust you?”: Explaining the predictions of any classifier’, in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. San Francisco, CA, August. ACM, pp. 1135–1144.

Salton, G. and Buckley, C. (1988) ‘Term-weighting approaches in automatic text retrieval’, *Information Processing & Management*, 24(5), pp. 513–523.

Shu, K., Sliva, A., Wang, S., Tang, J. and Liu, H. (2017) ‘Fake news detection on social media: A data mining perspective’, *ACM SIGKDD Explorations Newsletter*, 19(1), pp. 22–36.

Verma, P.K., Agrawal, P., Amorim, I. and Prodan, R. (2021) ‘WELFake: Word embedding over linguistic features for fake news detection’, *IEEE Transactions on Computational Social Systems*, 8(4), pp. 881–893.

Wang, W.Y. (2017) ‘“Liar, liar pants on fire”: A new benchmark dataset for fake news detection’, *arXiv preprint* arXiv:1705.00648.

Zhou, X. and Zafarani, R. (2020) ‘A survey of fake news: Fundamental theories, detection methods, and opportunities’, *ACM Computing Surveys*, 53(5), pp. 1–37.